ACTIVITY: CAMPUS SECURITY PATROL – PRIORITY-BASED ROUTE PLANNING

Subject: Design and Analysis of Algorithms (DAA) | Difficulty: Medium | Mode: Individual or pairs | By: Pradeep Shet

1. Scenario

The college is hosting a major event, and the campus security team needs to patrol the campus to ensure safety, especially during evening hours. The campus contains several buildings, facilities, gates and pathways. The team starts from the Main Gate and must visit important locations according to their assigned priority while following the available paths between buildings.

2. Campus Network Diagram



3. Campus Locations and Priorities

Location No.	Location	Priority
1	Library	2
2	Classrooms Block	3
3	Hostel	4
4	Sports Complex	1
5	Cafeteria	3
6	Admin Block	5
7	Science Block	2
8	Parking	4
9	Auditorium	5
10	Hospital	1

Priority rule: Priority 1 is the highest priority and Priority 5 is the lowest. When choosing among available, unvisited neighboring locations, visit the location with the highest priority first. If two locations have the same priority, choose the one with the smaller location number.

4. Important Rules

- Higher-priority locations must be visited earlier whenever they are available as unvisited neighbors.
- Do not visit a location more than once. Mark it as visited so that the patrol does not revisit it during backtracking.
- Follow only the direct paths shown in the campus network.
- Start the patrol at the Main Gate. For this activity, treat the Main Gate as the starting point labelled M and connect it to locations 8 (Parking) and 10 (Hospital).
- The team should record its decisions and explain how the priority rule affects the traversal.

5. What You Need to Do

Task 1 – Represent the Campus Network

Write the adjacency matrix for the campus locations using the connections shown in the network diagram. Use 1 if two locations are directly connected and 0 otherwise. Include the Main Gate as the starting location if your instructor asks you to represent it as an additional vertex.

Task 2 – Plan the Second Strategy

1. Start from the Main Gate.
2. Follow an available unvisited path as deeply as possible, choosing the highest-priority eligible neighbor first.
3. Record the visit order and show when the patrol backtracks to an earlier location.
4. Explain how marking a location as visited prevents repeated visits during backtracking.

Task 4 – Comparison and Analysis

- Which strategy reaches high-priority locations earlier under the stated neighbor-selection rule? Justify your answer.
- Which strategy explores a path deeply before returning?
- Which strategy is more suitable when the team wants to inspect nearby locations level by level?
- What changes if the Hospital (location 10) becomes disconnected from the network?

Task 5 – Final Recommendation (10 minutes)

Write a short recommendation to the campus security in-charge. Choose the more suitable traversal strategy for this patrol and justify your choice using safety priority, coverage, visit order and the need to avoid revisiting locations.

6. Submission Instructions

Complete the activity on one A4 sheet (both sides may be used). Present the adjacency matrix, traversal traces, queue/path states, analysis answers and recommendation neatly. Write the traversal order; show the steps used to obtain it.

**********Happy Travelling**********